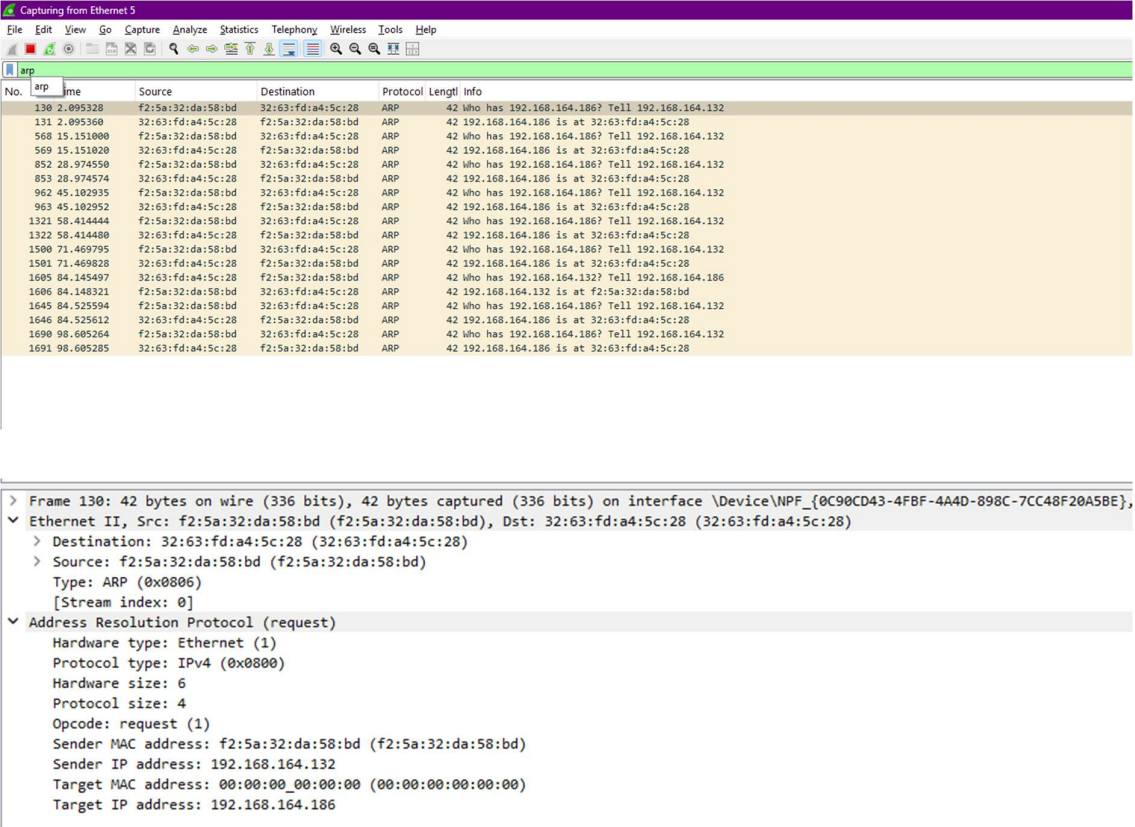


1. ARP Analysis

- The ARP request is broadcast to discover the MAC address associated with an IP.
- The sender includes both its IP (192.168.164.132) and MAC (f2:5a:32:da:58:bd) so the target knows where to send the reply.
- The reply contains the previously unknown MAC address, completing the address



Wireshark packet capture showing ARP traffic. The packet list shows frame 130 as an ARP request from 192.168.164.132 to 32:63:fd:a4:5c:28. The packet details pane shows the structure of the ARP request, including Ethernet II, Internet Protocol Version 4, and Address Resolution Protocol (request).

Frame 130: 42 bytes on wire (336 bits), 42 bytes captured (336 bits) on interface \Device\NPF_{0C90CD43-4F8F-4A4D-898C-7CC48F20A5BE}, Ethernet II, Src: f2:5a:32:da:58:bd (f2:5a:32:da:58:bd), Dst: 32:63:fd:a4:5c:28 (32:63:fd:a4:5c:28)

Destination: 32:63:fd:a4:5c:28 (32:63:fd:a4:5c:28)

Source: f2:5a:32:da:58:bd (f2:5a:32:da:58:bd)

Type: ARP (0x0806)

[Stream index: 0]

Address Resolution Protocol (request)

Hardware type: Ethernet (1)

Protocol type: IPv4 (0x0800)

Hardware size: 6

Protocol size: 4

Opcode: request (1)

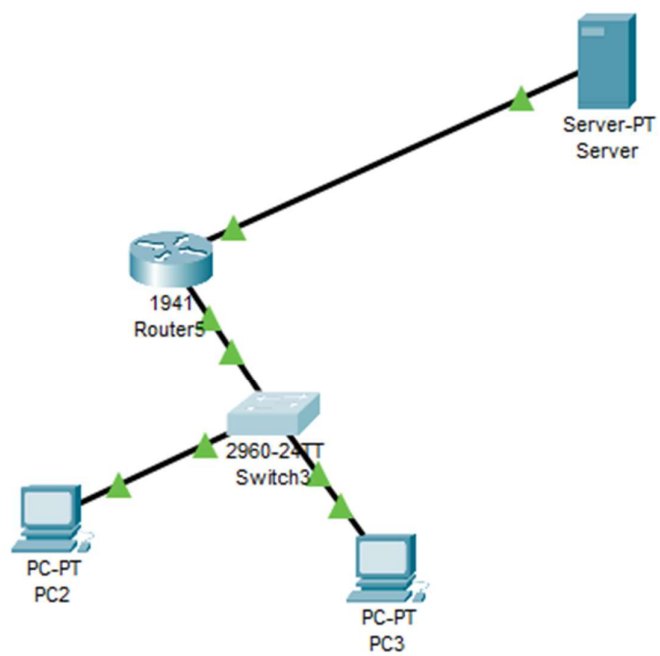
Sender MAC address: f2:5a:32:da:58:bd (f2:5a:32:da:58:bd)

Sender IP address: 192.168.164.132

Target MAC address: 00:00:00_00:00:00 (00:00:00:00:00:00)

Target IP address: 192.168.164.186

2. ARP Spoofing



```
C:\>ping 192.168.1.20

Pinging 192.168.1.20 with 32 bytes of data:

Reply from 192.168.1.20: bytes=32 time<1ms TTL=128
Reply from 192.168.1.20: bytes=32 time<1ms TTL=128
Reply from 192.168.1.20: bytes=32 time<1ms TTL=128
Reply from 192.168.1.20: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.20:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>arp -a

Internet Address      Physical Address      Type
192.168.1.20          000c.cf41.6694        dynamic

C:\>ping 192.168.1.1

Pinging 192.168.1.1 with 32 bytes of data:

Request timed out.
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>arp -a

Internet Address      Physical Address      Type
192.168.1.1           0060.70d2.1990        dynamic
192.168.1.20          000c.cf41.6694        dynamic

C:\>
```

PC3

Physical **Config** Desktop Programming Attributes

GLOBAL

Settings

Algorithm Settings

INTERFACE

FastEthernet0

Bluetooth

FastEthernet0

Port Status ☒ On

Bandwidth ☒ 100 Mbps ☐ 10 Mbps ☒ Auto

Duplex ☐ Half Duplex ☒ Full Duplex ☒ Auto

MAC Address 000C.CF41.6694

IP Configuration

☐ DHCP

☒ Static

IPv4 Address 192.168.1.20

Subnet Mask 255.255.255.0

IPv6 Configuration

☐ Automatic

☒ Static

IPv6 Address /

Link Local Address: FE80::20C:CFFF:FE41:6694

PC3

Physical

Config

Desktop

Programming

Attributes

GLOBAL

Settings

Algorithm Settings

INTERFACE

FastEthernet0

Bluetooth

FastEthernet0

Port Status

☒ On

Bandwidth

☒ 100 Mbps

☐ 10 Mbps

☒ Auto

Duplex

☐ Half Duplex

☒ Full Duplex

☒ Auto

MAC Address

0001.97C3.4602

IP Configuration

☐ DHCP

☒ Static

IPv4 Address

192.168.1.20

Subnet Mask

255.255.255.0

IPv6 Configuration

☐ Automatic

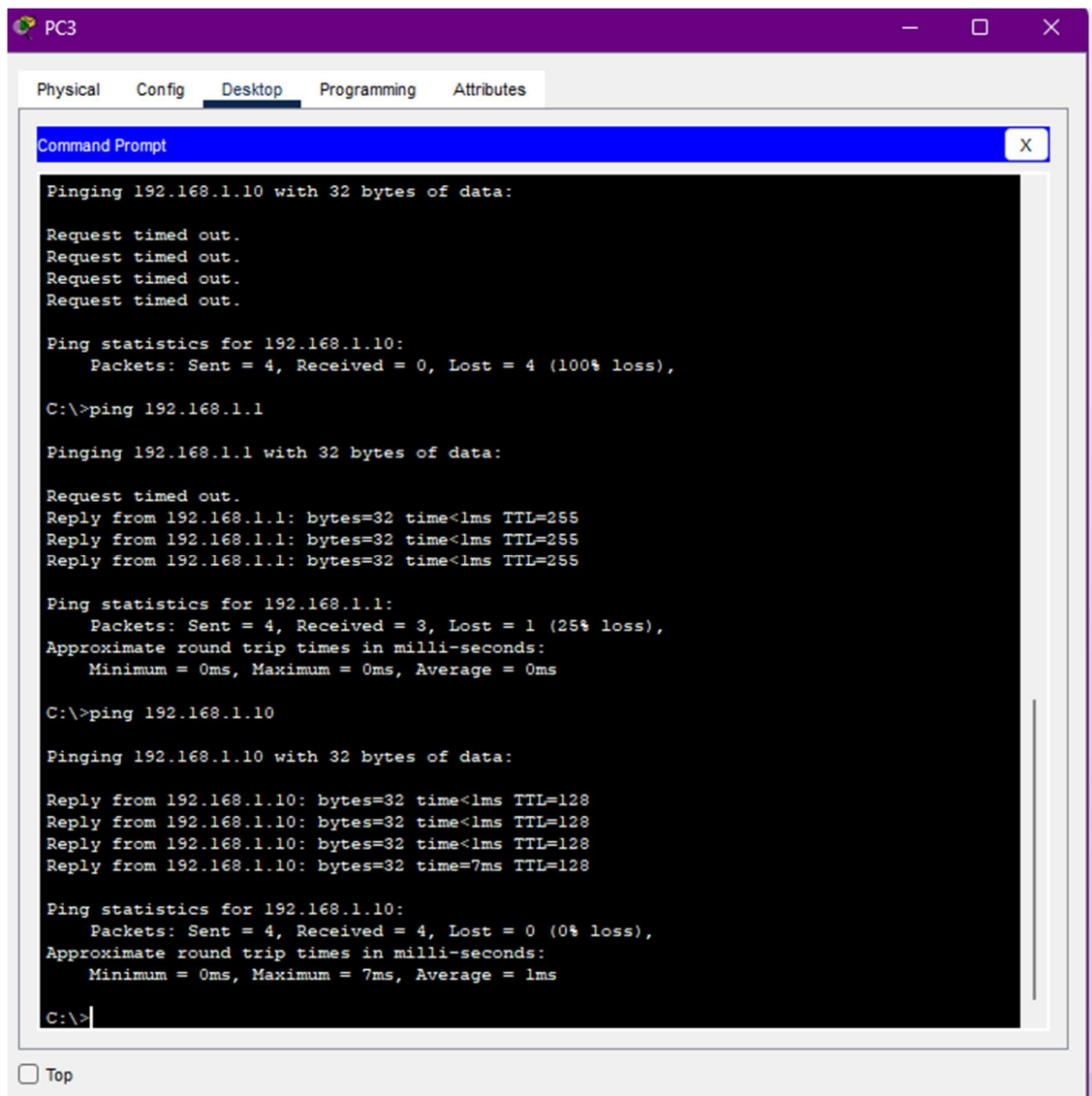
☒ Static

IPv6 Address

Link Local Address

FE80::20C:CFFF:FE41:6694

☐ Top



```
C:\>ping 192.168.1.1

Pinging 192.168.1.1 with 32 bytes of data:

Reply from 192.168.1.1: bytes=32 time=2ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 2ms, Average = 0ms

C:\>arp -a

Internet Address      Physical Address      Type
192.168.1.1           0001.97c3.4601       dynamic

C:\>
```

3. Manual Changing of IP Address

PC3

Physical Config Desktop Programming Attributes

IP Configuration X

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 192.168.1.20

Subnet Mask 255.255.255.0

Default Gateway 192.168.1.1

DNS Server 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address /

Link Local Address FE80::201:97FF:FEC3:4602

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

Authentication MD5

Username

Password

☐ Top

PC2

Physical Config **Desktop** Programming Attributes

IP Configuration X

Interface: FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address: 192.168.1.10

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.1.1

DNS Server: 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address: /

Link Local Address: FE80::260:5CFF:FE29:2D5A

Default Gateway:

DNS Server:

802.1X

☐ Use 802.1X Security

Authentication: MD5

Username:

Password:

☐ Top

```
C:\>ping 192.168.1.10

Pinging 192.168.1.10 with 32 bytes of data:

Reply from 192.168.1.10: bytes=32 time<1ms TTL=128
Reply from 192.168.1.10: bytes=32 time<1ms TTL=128
Reply from 192.168.1.10: bytes=32 time<1ms TTL=128
Reply from 192.168.1.10: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

4. DHCP Packet Capturing

Wireshark packet capture showing DHCP transactions. The packet list shows a DHCP Request, ACK, Discover, Offer, and ACK. The packet details for frame 79297 show the Ethernet II, Internet Protocol Version 4, and Dynamic Host Configuration Protocol (ACK) layers. The packet bytes pane shows the raw data in hexadecimal and ASCII.

5. Subnetting Task for 192.168.1.0/24 divided into 4 subnets

Subnet Mask Calculation

To divide a /24 network into 4 equal subnets:

- Original network: 192.168.1.0/24
- Number of subnets needed: 4
- Bits required for 4 subnets: 2 bits ($2^2 = 4$)
- New prefix length: $24 + 2 = /26$
- New subnet mask: 255.255.255.192

Subnet Ranges

Subnet	Network Address	First Usable IP	Last Usable IP	Broadcast Address
1	192.168.1.0	192.168.1.1	192.168.1.62	192.168.1.63
2	192.168.1.64	192.168.1.65	192.168.1.126	192.168.1.127
3	192.168.1.128	192.168.1.129	192.168.1.190	192.168.1.191

Subnet	Network Address	First Usable IP	Last Usable IP	Broadcast Address
4	192.168.1.192	192.168.1.193	192.168.1.254	192.168.1.255

Calculation Details

- Subnet mask in binary: 11111111.11111111.11111111.11000000
- Subnet mask in decimal: 255.255.255.192
- Number of hosts per subnet: $2^6 - 2 = 62$ hosts
- Subnet increment: 64 ($256 \div 4$)

Each subnet provides 62 usable addresses (excluding the network and broadcast addresses).

6. IP Address Classification

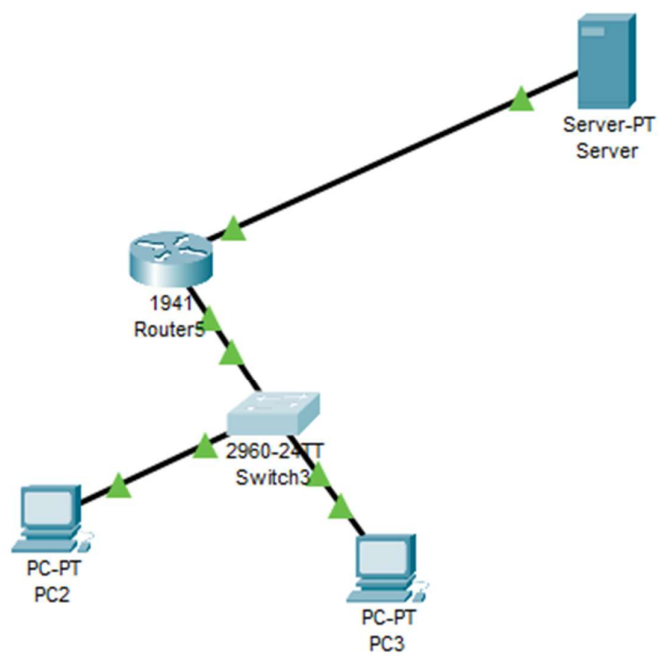
IP Address Classes for Given Addresses

IP Address	Class	Default Subnet Mask
10.1.1.1	Class A	255.0.0.0 (/8)
172.16.5.10	Class B	255.255.0.0 (/16)
192.168.1.5	Class C	255.255.255.0 (/24)

IP Address Ranges by Class

- Class A: 1.0.0.0 - 126.255.255.255
- Class B: 128.0.0.0 - 191.255.255.255
- Class C: 192.0.0.0 - 223.255.255.255

7. NAT



```
Router5
Physical Config CLI Attributes
IOS Command Line Interface

Press RETURN to get started!

Router>
Router>
Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#
Router(config)#
Router(config)#
Router(config)#interface GigabitEthernet0/0
Router(config-if)#ip address 192.168.1.1 255.255.255.0
Router(config-if)#ip nat inside
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

Router(config-if)#
Router(config-if)#
Router(config-if)#exit
Router(config)#
Router(config)#interface GigabitEthernet0/1
Router(config-if)#ip address 203.0.113.2 255.255.255.0
Router(config-if)#ip nat outside
Router(config-if)#
Router(config-if)#
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

Router(config-if)#
Router(config-if)#exit
```

Switch2

Physical Config CLI Attributes

IOS Command Line Interface

Press RETURN to get started!

%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/2, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up

Switch>
Switch>
Switch>
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#
Switch(config)#
Switch(config)#
Switch(config)#interface vlan 1
Switch(config-if)#ip address 192.168.1.1 255.255.255.0
Switch(config-if)#
Switch(config-if)#
Switch(config-if)#no shutdown

Switch(config-if)#
%LINK-5-CHANGED: Interface Vlan1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan1, changed state to up

Switch(config-if)#
Switch(config-if)#exit
Switch(config)#

☐ Top

```
IP NAT debugging is on
Router#
Router#
Router#
Router#
Router#show ip nat statistics
Total translations: 0 (0 static, 0 dynamic, 0 extended)
Outside Interfaces: GigabitEthernet0/1
Inside Interfaces: GigabitEthernet0/0
Hits: 3 Misses: 4
Expired translations: 4
Dynamic mappings:
Router#
```

```
Router#
Router#
Router#
NAT: s=192.168.1.20->203.0.113.2, d=203.0.113.10 [17]

NAT*: s=203.0.113.10, d=203.0.113.2->192.168.1.20 [4]

NAT: s=192.168.1.20->203.0.113.2, d=203.0.113.10 [18]

NAT*: s=203.0.113.10, d=203.0.113.2->192.168.1.20 [5]

NAT: s=192.168.1.20->203.0.113.2, d=203.0.113.10 [19]

NAT*: s=203.0.113.10, d=203.0.113.2->192.168.1.20 [6]

NAT: s=192.168.1.20->203.0.113.2, d=203.0.113.10 [20]

NAT*: s=203.0.113.10, d=203.0.113.2->192.168.1.20 [7]
```